$$x(t) \le \frac{\varepsilon t^2}{2} + \underbrace{\int_0^t s\cdot x(s)\,ds}_{\varphi(t)}, \qquad \varphi'(t) = t\,x(t) \le t\left(\frac{\varepsilon t^2}{2} + \varphi(t)\right) = t\,\varphi(t) + \frac{\varepsilon t^3}{2}$$

$$\varphi'(t) \le t\cdot\varphi(t) + \frac{\varepsilon t^3}{2}, \qquad \psi(t) = e^{t^2/2}\cdot\varphi(t)$$

$$\underbrace{t\cdot e^{\frac{t^2}{2}}\cdot\varphi(t)} + e^{\frac{t^2}{2}}\cdot\varphi'(t) \le \underbrace{t\cdot e^{t^2/2}\cdot\varphi(t)} + \frac{\varepsilon t^3}{2}$$

$$e^{\frac{t^2}{2}}\cdot\varphi'(t) \le \frac{\varepsilon t^3}{2} \implies \varphi'(t) \le \frac{\varepsilon t^3}{2}\cdot e^{-\frac{t^2}{2}} \implies \varphi(t) \le \underbrace{\varphi(0)}_{0} + \int_0^t \frac{\varepsilon s^3}{2}\cdot e^{-\frac{s^2}{2}}\,ds$$

$$\varphi(t) \le e^{\frac{t^2}{2}}\cdot\int_0^t \frac{\varepsilon s^3}{2}\cdot e^{-\frac{s^2}{2}}\,ds$$

$$x(t) \le \frac{\varepsilon}{2}\cdot\left(t^2 + e^{\frac{t^2}{2}}\cdot\int_0^t s^3\cdot e^{-\frac{s^2}{2}}\,ds\right)$$

$$\int_0^t s^3\cdot e^{-\frac{s^2}{2}}\,ds \underset{\downarrow}{=} \ldots = -e^{-\frac{t^2}{2}}(t^2+2)+2$$

$$\left[\begin{matrix} u = s^2 & du = 2s \\ dv = se^{-\frac{s^2}{2}} & v = -e^{-\frac{s^2}{2}}\end{matrix}\right]$$

A la manera del profesor:

$$-se^{-\frac{s^2}{2}} = \left(e^{-\frac{s^2}{2}}\right)'$$

$$\implies \int_0^t s^3\cdot e^{-\frac{s^2}{2}}\,ds = \int_0^t -s^2\left(e^{-\frac{s^2}{2}}\right)'ds = -\underbrace{\int_0^t \left(s^2\cdot e^{-\frac{s^2}{2}}\right)'ds}_{\to\ -t^2\cdot e^{-\frac{t^2}{2}}} + \underbrace{\int_0^t 2s\cdot e^{-\frac{s^2}{2}}\,ds}_{\to\ -2e^{-\frac{s^2}{2}}\Big]_0^t} = -e^{-\frac{t^2}{2}}\cdot(t^2+2)+2$$

16. Prove that, for every $\varepsilon \geq 0$, the problem

$$\begin{cases} u'(t) = t \sin u(t), \\ u(0) = \varepsilon, \end{cases}$$

has a unique solution globally defined in $[0, \infty)$, denoted by u_ε, and estimate $|u_\varepsilon(t) - u_0(t)|$ for every $t > 0$ and $\varepsilon > 0$.

$$\begin{cases} u'(t) = t \operatorname{sen} u(t) \\ u(0) = \varepsilon \ , \ \varepsilon > 0 \end{cases}$$

Definimos $f(t,u) = t \operatorname{sen} u \in C^\infty(\mathbb{R}\times\mathbb{R};\mathbb{R})$ y analítica en ambas variables. $\Rightarrow$ es localmente Lipschitziana

$\frac{\partial f}{\partial u}(t,u) = t\cos u \rightsquigarrow \left|\frac{\partial f}{\partial u}(t,u)\right| = |t\cos u| \leq |t| = t \leq \beta \quad \forall t \in [\alpha,\beta] \Rightarrow$ globalmente Lipschitz con respecto $u \in \mathbb{R}$

uniformemente en subintervalos compactos de $\mathbb{R}$. Aplicamos el Teorema de Cauchy - Lipschitz global en cada J_n

Sea $J_n = [0,n] \rightsquigarrow \bigcup_{n\geq 1} J_n = [0,\infty)$

$u_{n+1}\big|_{J_n} = u_n \Rightarrow \lim_{n\to\infty} u_n = u(t) \Rightarrow$ única solución en $[0,\infty)$

· Hacemos la estimación

$$u_\varepsilon(t) - u_0(t) = \varepsilon + \int_0^t (u_\varepsilon'(s) - u_0'(s))\,ds \quad \circledast$$

$$u_\varepsilon'(t) - u_0'(t) = t \operatorname{sen} u_\varepsilon(t) - t \operatorname{sen} u_0(t) = t(\operatorname{sen} u_\varepsilon(t) - \operatorname{sen} u_0(t)) = t\cos\xi(t)\cdot(u_\varepsilon(t) - u_0(t))$$

$$|u_\varepsilon'(t) - u_0'(t)| \leq t \cdot |u_\varepsilon(t) - u_0(t)|$$

$$\circledast \quad \underbrace{|u_\varepsilon(t) - u_0(t)|}_{x(t)} \leq \varepsilon + \int_0^t s \underbrace{|u_\varepsilon(s) - u_0(s)|}_{x(s)}\,ds$$

$$\varphi(t) = \int_0^t s\,x(s)\,ds \rightarrow \varphi'(t) = t\,x(t) \leq t(\varepsilon + \varphi(t)) \rightarrow \varphi'(t) \leq t\varphi(t) + t\varepsilon$$

$h' = th \rightarrow h(t) = e^{\frac{t^2}{2}}$, $\varphi(t) = e^{\frac{t^2}{2}}\cdot\psi(t)$ variamos coeficientes.

$$\underline{t\cdot e^{\frac{t^2}{2}}\cdot\psi(t)} + e^{\frac{t^2}{2}}\psi'(t) \leq \underline{t\cdot e^{\frac{t^2}{2}}\cdot\psi(t)} + t\varepsilon \rightarrow \psi'(t) \leq t\varepsilon e^{-\frac{t^2}{2}} \rightarrow$$

$$\psi(t) \leq \psi(t) + \int_0^t s\varepsilon e^{-\frac{s^2}{2}}\,ds \longrightarrow \psi(t) \leq \psi(0) - \left[\varepsilon e^{-\frac{s^2}{2}}\right]_0^t = \psi(0) - \varepsilon\cdot e^{-\frac{t^2}{2}} + \varepsilon = \varepsilon(1 - e^{\frac{t^2}{2}})$$

$$\psi(t) \leq \varepsilon\cdot(e^{\frac{t^2}{2}} - 1)$$

$$x(t) \leq \varepsilon + \varepsilon(e^{\frac{t^2}{2}} - 1) = \varepsilon e^{\frac{t^2}{2}}$$